

BAIO Podcast Scripts

Ready-to-record scripts · Narendra Sir

🎯 NEET & KCET Corner · 📖 Biology Stories · 🍳 Everyday Biology

🦇 Trap of the Week: Homologous vs Analogous

🎯 NEET & KCET Corner · II PUC · Ch 6 · Evolution · ~2 min

Narrator: *Welcome to NEET and KCET Corner on BAIO, with Narendra Sir. Today's trap: homologous versus analogous organs.*

Narendra Sir: Priya, a quick question. The wing of a bat and the wing of a butterfly — homologous or analogous?

Priya: Both are wings, Sir... so homologous?

Narendra Sir: That is exactly the trap! Many students lose a mark right here. Let's fix it forever.

Narendra Sir: Homologous organs have the same basic structure and the same origin, but they may do different jobs. Think of the forelimbs of a whale, a bat, a cheetah and a human.

Priya: Same bones — humerus, radius, ulna, carpals — but one swims, one flies, one runs and one writes!

Narendra Sir: Perfect. That shows divergent evolution: one ancestor, many directions.

Narendra Sir: Analogous organs do the same job but have different structures and origins. The wings of a butterfly and of a bird both help in flying, but they are built completely differently.

Priya: So that is convergent evolution — different ancestors arriving at the same solution.

Narendra Sir: Yes. My memory trick: Homologous — same Home, same origin. Analogous — same Action, different origin.

Priya: Home and Action! I'll never forget that, Sir.

Narendra Sir: Let's test you with exam favourites. The thorn of Bougainvillea and the tendril of Cucurbita?

Priya: Both are modified stems... so homologous.

Narendra Sir: Correct. The eye of an octopus and the eye of a mammal?

Priya: Same function, different origin — analogous.

Narendra Sir: The flippers of a penguin and a dolphin?

Priya: Both swim, but a penguin is a bird and a dolphin is a mammal — analogous!

Narendra Sir: Excellent. Sweet potato and potato?

Priya: Sweet potato is a root and potato is a stem. Same job, storage — so analogous.

Narendra Sir: You are thinking like a topper. Last one: the hearts and brains of vertebrates?

Priya: The same basic plan in all vertebrates — homologous.

Narendra Sir: Remember the rule: same structure, different function — homologous. Same function, different structure — analogous.

Narrator: *That's this week's trap. Now three quick questions — answer before you tap!*

🧠 **Quick quiz**

1. The wings of a butterfly and a bird are: — *Homologous organs / Analogous organs / Vestigial organs / Atavistic organs*
2. Homologous organs indicate: — *Convergent evolution / Divergent evolution / No evolution / Industrial melanism*
3. Sweet potato and potato are an example of: — *Homologous organs / Analogous organs / Vestigial organs / Adaptive radiation only*

Answers: 1-Analogous organs · 2-Divergent evolution · 3-Analogous organs

🌽 Trap of the Week: C3 vs C4 Plants

🎯 NEET & KCET Corner · Bonus · Photosynthesis in Higher Plants · ~2 min

Narrator: *Welcome back to NEET and KCET Corner. Today's trap: C3 versus C4 plants.*

Priya: Sir, I always mix them up. Isn't a C4 plant just a better C3 plant?

Narendra Sir: Not quite, Priya. The name comes from the first stable product of carbon fixation.

Narendra Sir: In C3 plants, the enzyme RuBisCO fixes carbon dioxide to RuBP, and the first stable product is 3-PGA — a three-carbon compound. Hence, C3.

Priya: And in C4 plants the first product has four carbons?

Narendra Sir: Yes — oxaloacetic acid, OAA. The enzyme is PEP carboxylase, and it works in the mesophyll cells.

Priya: Then where does the Calvin cycle happen in C4 plants?

Narendra Sir: In the bundle sheath cells. Large cells with many chloroplasts, thick walls and no intercellular spaces, arranged like a wreath around the veins. That is Kranz anatomy.

Priya: Kranz means wreath! That's how I'll remember it.

Narendra Sir: Now the trap NEET loves: do C4 plants have the Calvin cycle?

Priya: Hmm... I would have said no.

Narendra Sir: They do! The Calvin cycle runs in every photosynthetic plant. C4 plants simply add an extra step that concentrates carbon dioxide before it.

Priya: Why bother with that extra step?

Narendra Sir: To avoid photorespiration. At high temperatures RuBisCO starts grabbing oxygen instead of carbon dioxide and wastes energy. C4 plants pump so much carbon dioxide into the bundle sheath that RuBisCO stays busy with it.

Priya: So C4 plants do better in hot, sunny places.

Narendra Sir: Exactly — maize, sugarcane and sorghum. They tolerate higher temperatures, respond to high light and are more productive. Wheat and rice are C3 plants.

Priya: One more trap, Sir — the primary carbon dioxide acceptor?

Narendra Sir: In C3 plants it is RuBP, with five carbons. In C4 plants it is PEP — phosphoenolpyruvate — with three carbons.

Priya: So the acceptor has three carbons but the product has four. That's confusing!

Narendra Sir: That's why it's a trap. Say it aloud: PEP accepts, OAA appears.

Narrator: *Time for the quick quiz!*

🧠 **Quick quiz**

1. The first stable product of CO_2 fixation in C4 plants is: — *3-PGA / Oxaloacetic acid (OAA) / RuBP / PEP*
2. The primary CO_2 acceptor in C4 plants is: — *RuBP / PEP / OAA / PGA*
3. Kranz anatomy means: — *Large bundle sheath cells around the veins / Stomata only on the upper surface / No mesophyll cells / A thick cuticle*

Answers: 1-Oxaloacetic acid (OAA) · 2-PEP · 3-Large bundle sheath cells around the veins

⚡ KCET Rapid Fire: Human Health and Disease

🎯 NEET & KCET Corner · II PUC · Ch 7 · Human Health and Disease · ~2 min

Narrator: *It's KCET Rapid Fire! Ten quick questions on Human Health and Disease. After each question you get four seconds — answer aloud before Priya does!*

Narendra Sir: Number one. Which organism causes typhoid?

▮ [pause 4 s]

Priya: Salmonella typhi.

Narendra Sir: Correct — and typhoid is confirmed by the Widal test. Number two. Pneumonia is caused by?

▮ [pause 4 s]

Priya: Streptococcus pneumoniae and Haemophilus influenzae.

Narendra Sir: Three. Which stage of Plasmodium enters our body?

▮ [pause 4 s]

Priya: Sporozoites — through the bite of an infected female Anopheles mosquito.

Narendra Sir: Four. The chills and high fever of malaria are due to?

▮ [pause 4 s]

Priya: Haemozoin, a toxic substance released when the red blood cells burst.

Narendra Sir: Five. Amoebiasis is caused by?

▮ [pause 4 s]

Priya: Entamoeba histolytica. Houseflies act as mechanical carriers.

Narendra Sir: Six. Ringworm is caused by fungi of which genera?

▮ [pause 4 s]

Priya: Microsporum, Trichophyton and Epidermophyton.

Narendra Sir: Seven. Elephantiasis is caused by?

▮ [pause 4 s]

Priya: Wuchereria, the filarial worm, spread by female mosquitoes.

Narendra Sir: Eight. Which antibody is found in large amounts in colostrum?

▮ [pause 4 s]

Priya: IgA.

Narendra Sir: Nine. HIV mainly attacks which cells?

▮ [pause 4 s]

Priya: Helper T-lymphocytes. And macrophages act as an HIV factory.

Narendra Sir: Ten. Which test is widely used to detect HIV infection?

▮ [pause 4 s]

Priya: ELISA — enzyme-linked immunosorbent assay.

Narendra Sir: Ten out of ten if you answered with Priya! Write down the ones you missed and revise them tonight.

Narrator: *Now three bonus questions in the quiz.*

🧠 **Quick quiz**
1. The Widal test confirms: — *Malaria / Typhoid / Pneumonia / AIDS*
2. The chills and fever of malaria are due to: — *Sporozoites / Haemozoin / Gametocytes / Histamine*
3. ELISA is widely used to detect: — *Typhoid / HIV infection / Ringworm / Ascariasis*

Answers: 1-Typhoid · 2-Haemozoin · 3-HIV infection

⚖️ Assertion–Reason Made Easy

🎯 NEET & KCET Corner · II PUC · Ch 4 & 5 · Genetics · ~2 min

Narrator: *Assertion–Reason questions scare many students. Today Narendra Sir shows a simple three-step method.*

Priya: Sir, honestly, I read A and R and then just guess!

Narendra Sir: No more guessing. Step one: is the Assertion true? Step two: is the Reason true? Step three — only if both are true — does the Reason explain the Assertion? Join them with the word because.

Priya: Assertion because Reason. Let's try!

Narendra Sir: Assertion: the AB blood group shows codominance. Reason: both alleles, I A and I B, express themselves fully in the heterozygote.

Priya: Step one, true. Step two, true. Step three — AB shows codominance because both alleles express fully. Yes! Both are true and R explains A.

Narendra Sir: Perfect. Next. Assertion: in a dihybrid cross the F2 phenotypic ratio is 9 is to 3 is to 3 is to 1. Reason: the genes for seed colour and seed shape lie very close together on the same chromosome.

Priya: Assertion is true. But if the genes were close together, they would be linked and would not assort independently. So the Reason is false.

Narendra Sir: Correct — A is true, R is false. Next, from Molecular Basis of Inheritance. Assertion: tRNA is called an adaptor molecule. Reason: it has an anticodon that reads a codon on mRNA, and it also carries a specific amino acid.

Priya: Both true... and the Reason tells us why it's an adaptor. So R explains A.

Narendra Sir: Well done. Now a bonus trap — sometimes the Assertion is false and the Reason is true! Assertion: Mendel's law of dominance applies to every trait. Reason: in snapdragon, the F1 flowers are pink.

Priya: The Assertion is false — incomplete dominance exists. The Reason is true. So: A false, R true.

Narendra Sir: Exactly. Three steps every single time: true? true? because?

Priya: And step three only when both are true. Got it, Sir!

Narrator: *Quiz time — use the three steps!*

🧠 **Quick quiz**

1. If the Assertion is true and the Reason is false, choose: — *Both true, R explains A / Both true, R does not explain A / A true, R false / A false, R true*
2. Step three (does R explain A?) is checked only when: — *A is false / R is false / Both A and R are true / Neither is true*
3. tRNA is called an adaptor molecule because it: — *Carries the mRNA / Reads a codon and carries an amino acid / Forms the ribosome / Joins exons*

Answers: 1-A true, R false · 2-Both A and R are true · 3-Reads a codon and carries an amino acid

🫛 Mendel: The Monk Who Counted Peas

📖 Biology Stories · II PUC · Ch 4 · Principles of Inheritance and Variation · ~2 min

Narrator: *Brno, 1856. In a quiet monastery garden, a monk named Gregor Mendel is planting peas.*

Priya: Sir, why peas? Why not roses or mangoes?

Narendra Sir: Smart question, Priya. Peas grow fast, give many seeds, show clear contrasting traits, and can be self-pollinated or cross-pollinated by hand.

Narrator: *Mendel chose seven pairs of contrasting traits — tall and dwarf stems, round and wrinkled seeds, yellow and green seeds, violet and white flowers, and more.*

Gregor Mendel: First I need true-breeding plants — tall plants that always give tall plants, and dwarf plants that always give dwarf ones.

Narrator: *For seven years he crossed, counted and recorded — thousands and thousands of plants.*

Gregor Mendel: I crossed a tall plant with a dwarf one. Every plant in the first generation is tall! Where has the dwarf gone?

Narendra Sir: That is the F1 generation, Priya. The dwarf trait did not vanish — it was hiding.

Gregor Mendel: When I let the F1 plants self-pollinate, the dwarf plants came back — about one in every four!

Priya: The 3 to 1 ratio!

Narendra Sir: Yes. Mendel said each trait is controlled by a pair of factors — today we call them genes — and the pair separates when gametes form. That is the Law of Segregation.

Narrator: *In 1865 Mendel presented his work. Almost nobody understood it.*

Priya: Why, Sir? It sounds so clear now.

Narendra Sir: Communication was slow, using mathematics in biology was new, and nobody had seen the factors yet. People believed traits blend like paint, so separate, stable factors seemed strange.

Narrator: *Mendel died in 1884, still unknown as a scientist. Then in 1900, three scientists — de Vries, Correns and von Tschermak — independently rediscovered his results.*

Narendra Sir: Today, every Punnett square you draw is a thank-you note to a patient monk and his peas.

Priya: What is the lesson for us, Sir?

Narendra Sir: Count carefully, record honestly, and don't give up — even when nobody claps.

Narrator: *Now, three questions on Mendel's story.*

🧠 **Quick quiz**

1. Mendel did his experiments on: — *Drosophila / Garden pea / Snapdragon / Maize*
2. The F2 phenotypic ratio of a monohybrid cross is: — *1:2:1 / 3:1 / 9:3:3:1 / 1:1*
3. Mendel's work was rediscovered in: — *1865 / 1900 / 1884 / 1953*

Answers: 1-Garden pea · 2-3:1 · 3-1900

🧫 Fleming's Lucky Mould

📖 Biology Stories · II PUC · Ch 8 · Microbes in Human Welfare · ~2 min

Narrator: *London, 1928. The bacteriologist Alexander Fleming is working with culture plates of Staphylococci bacteria.*

Alexander Fleming: Hmm, one of these unwashed plates has gone mouldy. Wait — look at this!

Narrator: *Around the green mould, the bacteria could not grow. There was a clear ring where nothing lived.*

Priya: Sir, wouldn't most people just throw that plate away?

Narendra Sir: Most would! But Fleming asked: what is this mould making that stops bacteria?

Narrator: *The mould was Penicillium notatum. Fleming named the chemical penicillin.*

Narendra Sir: Penicillin is an antibiotic — a chemical made by one microbe that kills or slows down other disease-causing microbes.

Priya: So did penicillin save people right away?

Narendra Sir: Not yet. Its full potential as an effective antibiotic was established much later by Ernest Chain and Howard Florey.

Narrator: *During the Second World War, penicillin was used extensively to treat wounded soldiers.*

Narendra Sir: And in 1945, Fleming, Chain and Florey were awarded the Nobel Prize.

Priya: Sir, can antibiotics cure a cold?

Narendra Sir: Great question! No. The common cold is caused by a virus, and antibiotics act on bacteria. Taking antibiotics unnecessarily helps bacteria become resistant.

Narrator: *Today, antibiotics treat deadly diseases like plague, whooping cough, diphtheria and leprosy, which once killed millions.*

Narendra Sir: Fleming's discovery looked like luck. But luck only helped because his mind was ready to notice it.

Priya: So curiosity is the real discovery!

Narendra Sir: Exactly, Priya.

Narrator: *Quiz time!*

🧠 **Quick quiz**

1. Penicillin was discovered by: — *Louis Pasteur / Alexander Fleming / Robert Koch / Edward Jenner*
2. Penicillin was obtained from: — *Penicillium notatum / Streptococcus / Aspergillus niger / Trichoderma*
3. Antibiotics are NOT useful against: — *Bacterial pneumonia / The common cold virus / Plague / Diphtheria*

Answers: 1-Alexander Fleming · 2-Penicillium notatum · 3-The common cold virus

🩸 A Day in the Life of a Red Blood Cell

📖 Biology Stories · Bonus · Body Fluids and Circulation · ~2 min

Rubi the RBC: Hello! I'm Rubi, a red blood cell. I'm a tiny biconcave disc, and I have no nucleus — that leaves more room for haemoglobin.

Priya: How many of you are there, Rubi?

Rubi the RBC: About five to five and a half million in every cubic millimetre of your blood! We are made in the red bone marrow.

Narrator: *Rubi's day begins in the right side of the heart, carrying blood that has given away its oxygen.*

Rubi the RBC: The right ventricle pushes me through the pulmonary artery to the lungs. That is pulmonary circulation.

Narrator: *In the lungs, Rubi squeezes through a capillary wrapped around an alveolus.*

Rubi the RBC: Here oxygen rushes into me and binds to my haemoglobin — I become oxyhaemoglobin and turn bright red! Carbon dioxide leaves me.

Narendra Sir: Priya, why does oxygen move into the blood here?

Priya: Because the partial pressure of oxygen is higher in the alveoli than in the blood, so it diffuses in.

Narendra Sir: Exactly.

Rubi the RBC: Back to the left atrium through the pulmonary veins. Then the strong left ventricle launches me into the aorta — systemic circulation, my long journey to the tissues!

Narrator: *In a working muscle, Rubi meets low oxygen and lots of carbon dioxide.*

Rubi the RBC: I hand over my oxygen to the tissues. Some carbon dioxide rides on me as carbamino-haemoglobin, but most travels as bicarbonate, thanks to my enzyme, carbonic anhydrase.

Narendra Sir: Blood passes through the heart twice in one complete round. That is double circulation, Priya.

Priya: How long does Rubi live?

Rubi the RBC: About 120 days. Then I am destroyed in the spleen — the graveyard of RBCs. My iron is recycled to make new cells.

Narendra Sir: And new red cells are made all the time to replace them.

Rubi the RBC: So next time you climb the stairs and breathe hard, remember — millions of us are racing to bring you oxygen. Bye!

Narrator: *Now, three questions about Rubi's journey.*

🧠 **Quick quiz**

1. RBCs are destroyed mainly in the: — *Kidney / Spleen / Heart / Lungs*
2. The average lifespan of an RBC is about: — *20 days / 120 days / 1 year / 12 days*
3. Most CO_2 is carried in the blood as: — *Carbamino-haemoglobin / Bicarbonate / Dissolved CO_2 / Oxyhaemoglobin*

Answers: 1-Spleen · 2-120 days · 3-Bicarbonate

🌼 A Pollen Grain's Big Adventure

📖 Biology Stories · II PUC · Ch 1 · Sexual Reproduction in Flowering Plants · ~2 min

Polly the pollen grain: Hi! I'm Polly, a pollen grain. I was born inside an anther, in a pollen sac, from a microspore mother cell that divided by meiosis.

Narendra Sir: So Polly is haploid, Priya.

Polly the pollen grain: My coat has two layers — a tough exine made of sporopollenin, one of the most resistant organic materials known, and a thin intine. My exine has gaps called germ pores.

Priya: Polly, how do you travel?

Polly the pollen grain: My flower is pollinated by insects, so I'm a bit sticky. A bee lands to collect nectar, and I cling to its hairy body.

Narrator: *The bee flies to another flower of the same species.*

Polly the pollen grain: Landing! I'm on the stigma. Now the pistil checks me — am I the right kind of pollen? This is pollen–pistil interaction.

Narendra Sir: If Polly is compatible, the pistil accepts her. If not, she is rejected.

Polly the pollen grain: Accepted! I absorb water and push out a pollen tube through a germ pore. It grows down through the style towards the ovule.

Priya: What's inside the tube?

Polly the pollen grain: My two male gametes! The tube enters the ovule through the micropyle and then enters a synergid, guided by the filiform apparatus.

Narrator: *Now comes the most special event in flowering plants — double fertilisation.*

Polly the pollen grain: One male gamete fuses with the egg — syngamy — forming the diploid zygote. The other fuses with the two polar nuclei — triple fusion — forming the triploid primary endosperm nucleus.

Narendra Sir: Two fusions in one embryo sac — that is why it is called double fertilisation, and it is unique to flowering plants.

Narrator: *The zygote grows into the embryo, the endosperm nourishes it, the ovule becomes a seed and the ovary becomes a fruit.*

Polly the pollen grain: And somewhere in that seed, a new plant is waiting to grow — and one day make pollen like me!

Narrator: *Quiz time!*

🧠 **Quick quiz**

1. The exine of pollen is made of: — *Cellulose / Sporopollenin / Pectin / Lignin*
2. Triple fusion forms the: — *Zygote / Primary endosperm nucleus / Embryo / Synergid*
3. The pollen tube enters the embryo sac through a: — *Antipodal cell / Synergid / Egg cell / Chalaza*

Answers: 1-Sporopollenin · 2-Primary endosperm nucleus · 3-Synergid

🥛 Biology in Your Kitchen

🍳 Everyday Biology · II PUC · Ch 8 · Microbes in Human Welfare · ~2 min

Narrator: *Today we are in the kitchen with Narendra Sir and Priya, making breakfast.*

Priya: Sir, last night Amma added a spoon of curd to warm milk, and this morning it's all curd. How?

Narendra Sir: That spoon of curd is the inoculum. It carries millions of Lactobacillus and other lactic acid bacteria — LAB.

Narendra Sir: They grow in the milk and produce acids that coagulate and partially digest the milk proteins. That's curd!

Priya: Is curd healthier than milk?

Narendra Sir: Yes. LAB increase vitamin B12, and in our stomach they help keep disease-causing microbes in check.

Priya: And the idli batter that rose overnight?

Narendra Sir: Fermentation by bacteria. They release carbon dioxide gas, and the gas makes the batter puff up. That's why idli and dosa are soft and spongy.

Priya: Bread dough rises too.

Narendra Sir: That is baker's yeast — Saccharomyces cerevisiae. It ferments the sugars and releases carbon dioxide, and the dough rises.

Priya: Why do we keep the batter in a warm place?

Narendra Sir: Because these microbes work best when it is warm. In cold weather, fermentation is slow.

Priya: What about cheese with big holes?

Narendra Sir: Swiss cheese! The large holes are made by carbon dioxide from a bacterium called Propionibacterium sharmanii. Roquefort cheese is ripened by fungi.

Priya: So my breakfast is full of microbes!

Narendra Sir: Useful ones. Even toddy, the traditional drink made from palm sap, is made by fermentation. Microbes are tiny cooks working for us.

Narrator: *Next time you eat curd rice, thank the bacteria! Now the quiz.*

🧠 **Quick quiz**

1. Milk is converted into curd by: — *Yeast / Lactobacillus / Penicillium / Rhizobium*
2. Idli batter and bread dough puff up because of: — *Oxygen / Carbon dioxide / Nitrogen / Hydrogen*
3. The large holes in Swiss cheese are made by: — *Propionibacterium sharmanii / Aspergillus niger / Streptococcus / Trichoderma*

Answers: 1-Lactobacillus · 2-Carbon dioxide · 3-Propionibacterium sharmanii

🕵️ Myth or Fact? Health Edition

🍳 Everyday Biology · II PUC · Ch 7 · Human Health and Disease · ~2 min

Narrator: *It's Myth or Fact! Narendra Sir reads a statement, you decide in three seconds, then Priya answers.*

Narendra Sir: Statement one: antibiotics cure the common cold.

▮ [pause 3 s]

Priya: Myth! The common cold is caused by rhinoviruses, and antibiotics don't work on viruses.

Narendra Sir: Right. Statement two: malaria spreads from person to person by touch.

▮ [pause 3 s]

Priya: Myth. It spreads through the bite of an infected female Anopheles mosquito.

Narendra Sir: So what is the best prevention?

Priya: Mosquito nets, repellents, no stagnant water around the house — and fish like Gambusia that eat mosquito larvae.

Narendra Sir: Excellent — that is in your textbook too. Statement three: vaccines give you the disease.

▮ [pause 3 s]

Priya: Myth. Vaccines contain inactivated or weakened pathogens, or their proteins. They train memory cells without causing the disease.

Narendra Sir: Statement four: AIDS spreads by shaking hands or sharing food.

▮ [pause 3 s]

Priya: Myth! HIV spreads through sexual contact with an infected person, transfusion of contaminated blood, sharing infected needles, and from an infected mother to her child.

Narendra Sir: Very important. People living with HIV should never be isolated. Statement five: smoking harms only the lungs.

▮ [pause 3 s]

Priya: Myth. Nicotine raises blood pressure and heart rate, and smoking is linked to cancers, bronchitis and emphysema.

Narendra Sir: And smoking increases carbon monoxide in the blood, which reduces the oxygen it can carry. Statement six: adolescence is the most vulnerable time to start using drugs.

▮ [pause 3 s]

Priya: Fact. Curiosity, peer pressure and stress make adolescents vulnerable.

Narendra Sir: Yes — and talking to parents and teachers, and getting help early, makes all the difference.

Narrator: *Five myths busted and one fact confirmed! Now the quiz.*

🧠 **Quick quiz**

1. The common cold is caused by: — *Bacteria / Rhinoviruses / Fungi / Protozoa*
2. Gambusia fish help control malaria because they: — *Kill bacteria / Eat mosquito larvae / Fix nitrogen / Make vaccines*
3. HIV does NOT spread by: — *Contaminated blood / Sharing needles / Shaking hands / Infected mother to child*

Answers: 1-Rhinoviruses · 2-Eat mosquito larvae · 3-Shaking hands

🍌 Why Do Bananas Ripen Faster Together?

🍳 Everyday Biology · Bonus · Plant Growth and Development · ~2 min

Narrator: *Priya kept some raw bananas in a bag with one ripe banana. By morning, all of them were yellow!*

Priya: Sir, is it magic?

Narendra Sir: It's a gas — ethylene. Ripening fruits release ethylene, a plant hormone, and it spreads to the fruits around them.

Priya: A hormone that is a gas!

Narendra Sir: Yes, a simple gaseous plant growth regulator. It hastens fruit ripening, and it raises the rate of respiration during ripening — called the respiratory climacteric.

Priya: Do farmers use it?

Narendra Sir: Yes, through ethephon, which releases ethylene. It hastens ripening in tomatoes and apples, and promotes female flowers in cucumbers, which increases the yield.

Priya: What other plant hormones are there?

Narendra Sir: Auxins, like IAA, help stem cuttings form roots and cause apical dominance. Gibberellins lengthen stems — sprayed on sugarcane they increase the yield, and on grapes they lengthen the stalks.

Priya: And cytokinins?

Narendra Sir: They promote cell division and delay leaf senescence, keeping leaves young for longer.

Narendra Sir: Abscisic acid, ABA, is the stress hormone. It closes the stomata during drought and keeps seeds dormant.

Priya: So ABA is the stop hormone, and most of the others are go hormones!

Narendra Sir: Nice way to put it. Ethylene also breaks seed and bud dormancy and starts germination in peanut seeds.

Narrator: *So, to ripen raw mangoes faster, keep them in a paper bag with a ripe banana.*

Priya: And to keep them raw for longer?

Narendra Sir: Keep them cool and away from ripe fruits — less ethylene, slower ripening.

Narrator: *Quiz time!*

🧠 **Quick quiz**

1. The gaseous plant hormone is: — *Auxin / Ethylene / Gibberellin / Cytokinin*
2. The hormone that delays leaf senescence is: — *ABA / Cytokinin / Ethylene / Auxin*
3. The stress hormone that closes stomata is: — *Abscisic acid / IAA / Gibberellin / Ethephon*

Answers: 1-Ethylene · 2-Cytokinin · 3-Abscisic acid

🐍 Karnataka's Green Treasure: the Western Ghats

🍳 Everyday Biology · II PUC · Ch 13 · Biodiversity and Conservation · ~2 min

Narrator: *Today Narendra Sir takes Priya on a trip to the Western Ghats of Karnataka.*

Narendra Sir: Priya, the Western Ghats together with Sri Lanka form one of the world's biodiversity hotspots.

Priya: What makes a place a hotspot?

Narendra Sir: Very high species richness, high endemism — species found nowhere else — and a high threat to those species.

Narendra Sir: There are 34 hotspots in the world. Together they cover less than two percent of the earth's land area, yet hold a huge share of its species.

Priya: Three of them are in India, right?

Narendra Sir: Yes — the Western Ghats and Sri Lanka, Indo-Burma, and the Himalaya.

Narrator: *In the rainforests of Kudremukh and Agumbe, frogs, orchids and birds are everywhere.*

Priya: Agumbe is famous for the king cobra!

Narendra Sir: Right. And the lion-tailed macaque, an endangered monkey, is found only in the Western Ghats.

Priya: Why are these species threatened?

Narendra Sir: Your textbook calls the causes the Evil Quartet: habitat loss and fragmentation, over-exploitation, alien species invasions and co-extinctions. Habitat loss is the biggest.

Priya: How do we protect them?

Narendra Sir: In situ conservation protects species in their natural home — national parks, wildlife sanctuaries and biosphere reserves, like Bandipur and Nagarhole in Karnataka.

Narendra Sir: Ex situ conservation protects them outside — in zoos, botanical gardens and seed banks, and by cryopreserving gametes.

Priya: And sacred groves?

Narendra Sir: Yes! Forest patches protected by communities for religious reasons. The Western Ghat regions of Karnataka have many — the devarakadu — where rare plants survive.

Narrator: *So next time you visit the Ghats, remember — you are walking through one of the earth's richest treasure chests.*

Priya: And it's our job to protect it!

Narrator: *Quiz time!*

🧠 **Quick quiz**

1. How many biodiversity hotspots does India have? — *1 / 3 / 5 / 34*
2. The biggest cause of biodiversity loss is: — *Alien species / Habitat loss and fragmentation / Co-extinction / Over-exploitation*
3. National parks are an example of: — *Ex situ conservation / In situ conservation / Cryopreservation / Seed banks*

Answers: 1-3 · 2-Habitat loss and fragmentation · 3-In situ conservation